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Feature structure co-optimized augmented network for graph anomaly detection

Yingyue Zhang, Huifang Ma *, Rui Bing, Meihuizi Jia, Xiaofei Wang

College of Artificial intelligence and Computer Science, Northwest Normal University, 730070, Lanzhou Gansu, China

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ABSTRACT

Graph Anomaly Detection (GAD) aims to identify nodes that deviate from attribute or structural norms. Despite recent advances, redundant features and structural noise continue to obscure anomaly signals and bias node representations. We propose FSCAN (Feature-Structure Co-optimized Augmented Network), a plug-in module that jointly optimizes feature selection and graph refinement. FSCAN leverages spectral contribution-based selection, grounded in the Spiked Covariance Model and Singular Value Decomposition, to identify label-relevant features, while its adaptive refinement prunes noisy edges from head nodes and augments connections for tail nodes to enhance information propagation. Experiments on benchmark datasets show that FSCAN outperforms state-of-the-art baselines. Specifically, on the Questions dataset, when integrated with a supervised backbone, FSCAN achieves a GMean of 67.35%, representing a 6.63% improvement over the best baseline; when integrated with an unsupervised backbone, the GMean reaches 22.49%, corresponding to a 10.98% improvement over the baseline. Our code is publicly available at https://github.com/mala-tang/FSCAN.

1. Introduction

Graph Anomaly Detection (GAD) plays a crucial role in identifying abnormal patterns in network data, with broad applications across domains including social fraud detection (Zhang et al., 2025, 2024) and financial transaction monitoring (Xu et al., 2024). Recent GAD approaches typically employ Graph Neural Networks (GNNs) within encoder-decoder frameworks. However, the complexity of real-world graphs presents two key challenges: (1) **redundant features dilute anomaly signals**, weakening representational capacity; and (2) **noisy edges obscure structural patterns**, hindering effective aggregation. These challenges degrade the performance of GNNs in anomaly detection tasks. Consequently, previous works have pursued two directions: feature optimization and structure refinement.

Feature optimization approaches in GAD predominantly fall into two main paradigms: constraint-based disentanglement and group-wise aggregation. Specifically, GDN (Gao et al., 2023b) implements the former by systematically separating invariant and contextual features via class and connectivity constraints, while BioGNN (Gao et al., 2024) adopts the latter, employing group-wise message passing mechanisms to enhance anomaly signal capture. Despite their demonstrated effectiveness, both methods suffer from a limitation: they rely on high-dimensional raw features and insufficiently address the adverse effects of feature redundancy.

* Corresponding author.
E-mail addresses: 2022212152@nwnu.edu.cn (Y. Zhang), mahuifang@nwnu.edu.cn (H. Ma), bingrui@nwnu.edu.cn (R. Bing), jiameihuizi@nwnu.edu.cn (M. Jia), 202421162207@nwnu.edu.cn (X. Wang).

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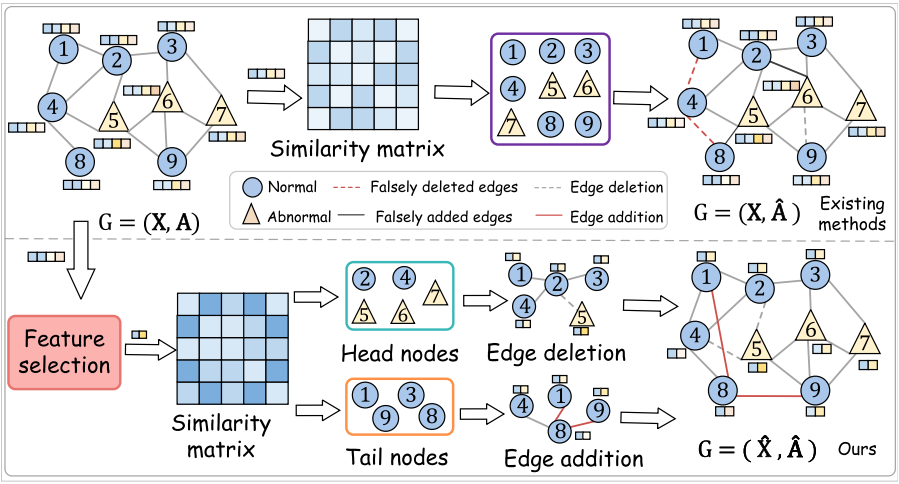


Fig. 1. The difference between existing methods and ours.

Structure optimization approaches generally adopt a global unified strategy–removing heterophilous edges or adding homophilous ones. GHRN (Gao et al., 2023a) leverages graph spectral theory to filter noisy edges, TAM (Qiao & Pang, 2023) prunes edges based on one-class homophily, and SparseGAD (Gong et al., 2023) integrates both edge deletion and addition strategies for structure refinement. However, these approaches insufficiently account for the inherent differences between head and tail nodes: as shown in Fig. 1 (top), head nodes tend to form misleading heterophilous links, while tail nodes risk losing crucial homophilous connections. Although AD-GCL (Xu et al., 2025b) distinguishes between head and tail nodes, it focuses on using head-node knowledge to guide pseudo-tail generation, without optimizing head nodes directly. Furthermore, most methods determine edge types based on similarity in the original feature space, where redundancy frequently leads to false homophilous links and removal of true ones, exacerbating feature-structure noise entanglement.

To address the aforementioned challenges, we conduct detailed data analysis and reveal two key empirical observations. **Observation 1: Redundant features obscure abnormal signals.** Redundant feature dimensions hinder the model's ability to capture meaningful patterns, diluting the signals that differentiate abnormal nodes. To tackle this issue, we propose a spectral contribution-based feature selection mechanism (Fig. 1 (bottom)), which leverages the spectral contribution of feature to identify and eliminate uninformative dimensions. Consequently, the model's sensitivity to anomalies is significantly enhanced. **Observation 2: Head-tail structural disparities affect classification performance.** Real-world graphs typically follow a power-law distribution, resulting in a minority head (high-degree) nodes and a majority tail (low-degree) nodes. This imbalance introduces distinct challenges: tail nodes, due to sparse connectivity, are prone to cumulative aggregation errors, while head nodes, with dense and diverse neighborhoods, exhibit higher misclassification rates. To mitigate these challenges, we introduce a structure-aware optimization strategy. For head nodes, we dynamically prune low-similarity edges based on feature similarity to suppress noise propagation. For tail nodes, we enhance local connectivity by establishing edges to their most similar non-neighbor nodes, thereby improving structural robustness (Fig. 1 (bottom)). Based on these observations, we propose FSCAN, which jointly selects features and refines structures to improve anomaly detection performance.

Our main contributions are summarized as follows:

- **Model Innovation:** We propose a feature-structure co-optimized framework that enhances anomaly separability through informative feature selection and node-aware structural refinement based on local density.
- **Theoretical Innovation:** We provide the first theoretical analysis quantifying the correlation between Singular Value Decomposition (SVD) derived features and node labels, offering new insights into spectral-based node anomaly detection methods.
- **Empirical Validation:** Extensive experiments on real-world GAD benchmarks show that our method consistently outperforms strong baselines, validating its effectiveness.

2. Related works

2.1. Feature selection

High-dimensional data often contain redundant and irrelevant features, which not only increase computational cost but also degrade model performance. Feature selection, as an effective solution to this problem, aims to select the most informative subset of features to improve learning efficiency and accuracy. Depending on the selection strategy, existing methods can be broadly categorized into three types. Filter methods (Buchaiah & Shakya, 2022a; Gong et al., 2024) evaluate feature importance using statistical criteria (e.g., variance, mutual information) independently of the model, offering high efficiency. Wrapper methods (Chaudhuri, 2024; Hassan et al., 2025) rely on specific learning models to evaluate subsets, often achieving better performance but with high computational cost.

Embedded methods (Xu et al., 2025a) integrate feature selection into model training process, balancing efficiency and adaptability. High-dimensional redundancy from rich node attributes can mask anomalies and impair GNN performance, making feature selection crucial for GAD.

2.2. Graph structure learning (GSL)

Existing GSL (Li et al., 2023; Zhou et al., 2023) can be broadly categorized into two types: for homogeneous graphs and for heterogeneous graphs. Homogeneous GSL assumes uniform node and edge types and can be further divided into two subgroups: topology reconstruction-based methods (Li et al., 2022; Liu et al., 2022a; Wang et al., 2021; Zhao et al., 2023), which aim to optimize graph structure, and topology initialization-based methods, such as NodeFormer (Liu et al., 2022c; Wu et al., 2024, 2022), which focus on constructing initial graph topology. Heterogeneous GSL targets graphs with multiple types of nodes and edges, aiming to capture more complex interactions (Bi et al., 2024; Zhao et al., 2021). In GAD, GSL methods are typically classified by spatial-based techniques (e.g., TAM, SparseGAD) and spectral-based approaches (e.g., GHRN). While these methods improve GAD performance, most rely on globally uniform operations (e.g., shared filters or edge weighting) and overlook structural differences among nodes.

3. Preliminaries

In this section, we define the key notations and problem formulation. We also conduct a detailed analysis of the feature and structural characteristics of GAD datasets to better understand how they influence the detection performance.

3.1. Notations and problem formulation

Following general settings, the object of graph anomaly detection is an attributed graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$ is the node set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set. The adjacency matrix $\mathbf{A} \in \{0,1\}^{n \times n}$ satisfies $\mathbf{A}_{i,j} = 1$ if $(v_i, v_j) \in \mathcal{E}$, otherwise $\mathbf{A}_{i,j} = 0$. Node attributes are represented by the feature matrix $\mathbf{X} \in R^{n \times f}$. A subset $\mathcal{V}_L \subset \mathcal{V}$ has binary labels $\mathbf{y}_L \in \{0,1\}$ (0 for normal, 1 for abnormal), while labels $\mathbf{y}_U$ for other nodes $\mathcal{V}_U \subset \mathcal{V} \setminus \mathcal{V}_L$ remain unknown. As per the setup in Liu et al. (2021), given a threshold K, tail nodes are defined as $\mathcal{V}_{tail} = \{v : |\mathcal{N}_v| \leq K\}$, while the remaining nodes are head nodes $\mathcal{V}_{head} = \{u : |\mathcal{N}_u| > K\}$. Notably, $K = \mu + \sigma$ is set to the sum of the dataset's mean degree and standard deviation.

The objective of this work is to optimize the original feature matrix $\mathbf{X}$ and adjacency matrix $\mathbf{A}$, and feed the optimized $\hat{\mathbf{X}}$ and $\hat{\mathbf{A}}$ into an anomaly detection model $f(\cdot)$ to enhance anomaly detection performance.

3.2. Data analysis

This subsection analyzes the impact of feature redundancy and structural differences on GAD performance using the Questions and BlogCatalog, offering theoretical and empirical support for subsequent method design.

3.2.1. Feature redundancy analysis

To investigate the impact of feature redundancy on GAD, we analyze the spectral energy distributions of normal and abnormal nodes, with the results presented in Fig. 2. The features used in this analysis fall into three categories: raw features; features selected by TFI (Zheng et al., 2025), which is specifically designed for graph-structured data; and the features selected by our proposed method based on Spectral Contribution (SC).

Following the energy ratio definition in GAD (Tang et al., 2022), the spectral energy ratio of a feature $\mathbf{x}_i$ is given by $\frac{\hat{\mathbf{x}}_i^2}{\sum_i^n \hat{\mathbf{x}}_i^2}$, where $\hat{\mathbf{x}}_i = \mathbf{U}_L^{\mathrm{T}} \mathbf{x}_i$ is the graph fourier transform of $\mathbf{x}_i$, and n is the number of nodes. In this context, low-frequency concentration indicates that features are smooth over the graph, i.e., similar across neighboring nodes, while high-frequency concentration reflects local differences. In anomaly detection, we expect normal nodes to exhibit low-frequency dominance due to structural consistency, while abnormal nodes should show more high-frequency components, revealing local inconsistency. Feature selection should therefore enhance frequency-domain separability, making normal and abnormal nodes more distinguishable in the spectral space.

As shown in Fig. 2, the raw feature space often exhibits spectral homogeneity–normal and abnormal nodes may both concentrate in either low or high-frequency bands. This suggests significant redundancy in raw features, which masks the spectral differences between node types and weakens their discriminability. On the Questions, both selection methods have limited impact on the spectral distribution of normal nodes, but SC notably increases the high-frequency ratio of anomalous nodes, enhancing their distinction from neighbors. On the BlogCatalog, both methods improve the low-frequency concentration of normal nodes, but SC shows a much stronger effect than TFI. These results demonstrate that SC effectively removes redundancy based on dataset characteristics and improves the spectral separability between normal and abnormal nodes.

3.2.2. Head-tail node structural differentiation analysis

Most existing GAD methods adopt a unified global edge processing strategy, which implicitly assumes a uniform degree distribution. However, real-world graphs typically follow a power-law distribution, resulting in significant structural and performance differences between head and tail nodes. To systematically assess the impact of such differences on anomaly detection, Fig. 3 presents the degree distributions and the number and proportion of misclassified nodes across degree intervals on the test set of two datasets.

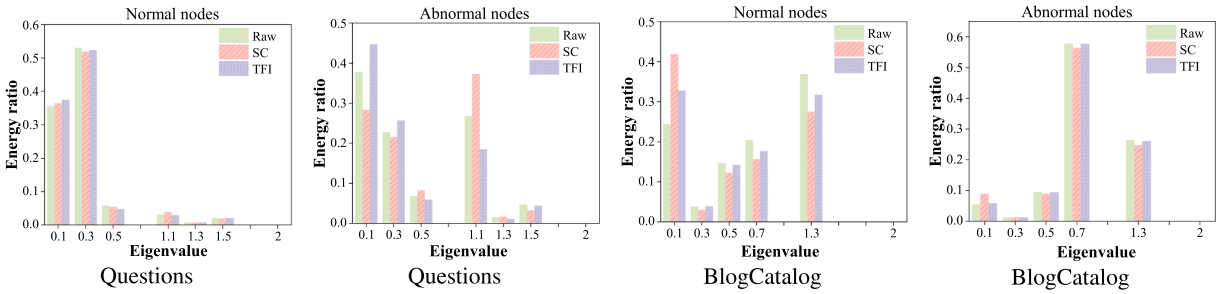


Fig. 2. Spectral energy distribution of normal and abnormal nodes under different feature selection strategies.

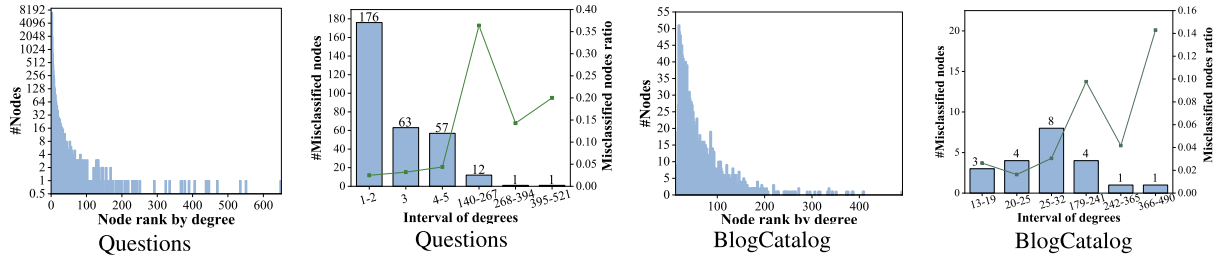


Fig. 3. Degree distribution and misclassification of head and tail nodes on the test sets of Questions and BlogCatalog.

As shown in Fig. 3, data exhibits a typical power-law distribution–tail nodes are abundant, while head nodes are scarce. Furthermore, Fig. 3 also reveals clear differences in misclassification patterns: tail nodes, due to their large population, contribute more to the absolute number of errors, yet their interval-wise misclassification rate is low; head nodes, though fewer in number, show a significantly higher misclassification rate within their degree range. These discrepancies pose a major challenge to detection performance. A unified edge processing strategy struggles to balance both ends.

We argue that head nodes, with dense and complex neighborhoods, are more susceptible to structural noise introduced by redundant edges, and thus benefit more from edge pruning. In contrast, tail nodes suffer from sparse neighborhoods and limited information propagation, making edge augmentation more beneficial for enhancing connectivity.

4. Methodology

Based on the preceding analysis, we introduce FSCAN to enhance GAD task, whose overall framework is illustrated in Fig. 4. In the feature stage, we theoretically analyze the relationship between features and labels, based on which we design a spectral contribution-based feature selection to eliminate redundant features that may hinder detection performance. In the structure stage, FSCAN adopts differentiated strategies for head and tail nodes to better fit their local structural characteristics. It prunes noisy edges for head nodes and adds edges for tail nodes to match their structural needs. This helps reduce noise and improve node representations. Finally, the optimized features and structure are fed into a GNN for anomaly detection.

4.1. Feature selection via spectral contribution

In GAD, identifying informative features is critical for distinguishing anomalous nodes, especially when anomalies manifest as subtle deviations. Although features highly correlated with node labels are intuitively more effective, relying on label information for feature selection risks label leakage. Therefore, an unsupervised approach is preferred. To this end, we propose a spectral contribution-based feature selection method. Leveraging the Spiked Covariance Model (SCM) (Ding & Yang, 2021) and Singular Value Decomposition (SVD), we theoretically demonstrate that, under certain statistical assumptions, features with higher contributions to the right singular vectors tend to be more correlated with the underlying label structure. This insight allows us to select informative features purely based on their spectral contributions, without requiring access to any label information, thereby avoiding potential label leakage.

4.1.1. Theoretical derivation

Let graph contain n nodes and f features, with the node feature matrix denoted as $\mathbf{X} \in R^{n \times f}$, where each row represents a node and each column corresponds to a feature. The node labels are given by $\mathbf{y} \in \{\pm 1\}^n$. Features are generated following the SCM:

$$\mathbf{X} = \sqrt{\frac{\eta}{n}}\mathbf{y}\mathbf{u}^T + \frac{\mathbf{Z}}{\sqrt{f}}, \tag{1}$$

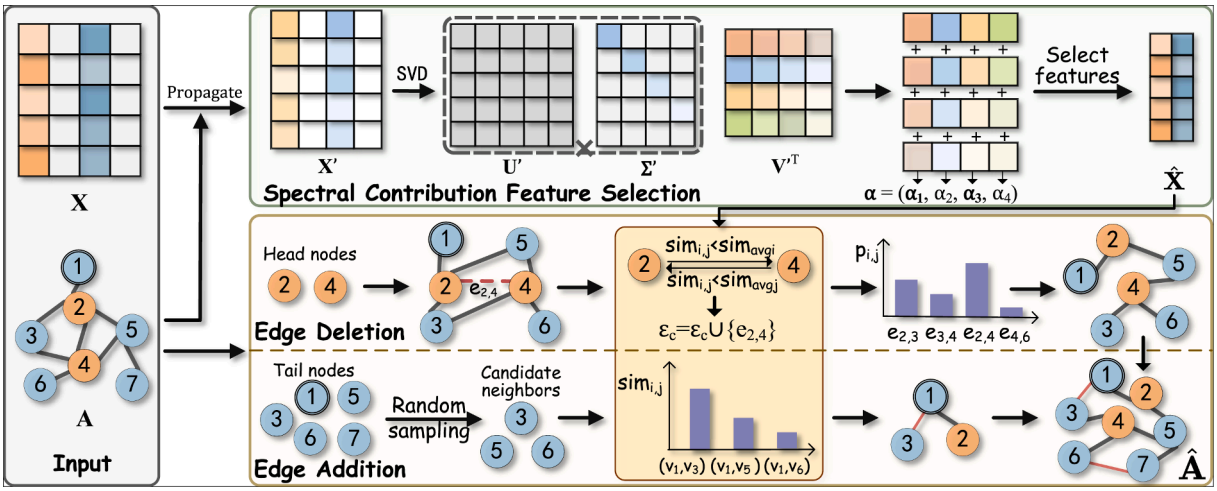


Fig. 4. Framework of FSCAN.

where, η controls signal to noise ratio, $\mathbf{u} \sim \mathcal{N}\left(0, \mathbf{I}_\mathbf{f}/f\right)$ is the direction vector, and $\mathbf{Z} \in R^{n\times f}$ is gaussian noise.

Applying SVD to $\mathbf{X}$, we have $\mathbf{X} = \mathbf{U}_\mathbf{X}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{V} = \left[\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_f\right] \in R^{f\times f}$ is the right singular matrix. Each v_j represents an orthogonal basis vector in the feature space, forming a complete set of feature bases. Normal and abnormal features can both be linearly represented by this basis.

Under strong signal conditions ($\eta \gg \frac{1}{\sqrt{f}}$) and in the high-dimensional limit ($n, f \to \infty$), random matrix theory (Benaych-Georges & Nadakuditi, 2012) shows that the top singular value separates from the bulk spectrum (i.e., spectral separation occurs), and the first right singular vector $\mathbf{v}_1$ aligns with the true signal direction $\mathbf{u}$, i.e., $\mathbf{v}_1 \approx \mathbf{u}$.

Let $\mathbf{x}_i = \mathbf{X}[:, i] \in \mathbb{R}^n$ be the ith feature vector. Its covariance with the label vector $\mathbf{y}$ is:

$$\mathbb{E}[\mathbf{x}_i^T\mathbf{y}] = \sqrt{\frac{\eta}{n}}\mathbf{u}_i \cdot \mathbb{E}[\mathbf{y}^T\mathbf{y}] + \frac{1}{\sqrt{f}}\mathbb{E}[\mathbf{Z}_i^T\mathbf{y}], \tag{2}$$

where $\mathbf{Z}_i \in \mathbb{R}^n$ denotes the ith column of $\mathbf{Z}$. Since $\mathbf{Z}_i$ and $\mathbf{y}$ are independent, the second term vanishes. Moreover, because $\mathbf{y} \in \{\pm 1\}^n$, we have $\mathbb{E}[\mathbf{y}^T\mathbf{y}] = n$. Thus,

$$\mathbb{E}[\mathbf{x}_i^T\mathbf{y}] = \sqrt{\eta n}\,\mathbf{u}_i, \tag{3}$$

indicating that the covariance (and hence correlation after normalization) between $\mathbf{x}_i$ and $\mathbf{y}$ is proportional to $\mathbf{u}_i$. Given $\mathbf{v}_1 \approx \mathbf{u}$, we conclude:

$$Corr(\mathbf{x}_i, \mathbf{y}) \propto \mathbf{v}_{1,i}. \tag{4}$$

This suggests that the larger the ith entry in the dominant right singular vector, the stronger the discriminative power of feature i.

However, in graph anomaly detection, both the strong-signal assumption (i.e., dominance of a single principal direction) and the high-dimensional limit often fail to hold. As a result, the correlation between features and labels is not necessarily concentrated in the first singular vector $\mathbf{v}_1$. Considering that normal nodes tend to align with neighbors while anomalous nodes often deviate across multiple components, we amplify these differences by squaring the entries of the right singular vectors. Therefore, we define the spectral contribution of feature i as:

$$\alpha_i = \sum_{j=1}^{r} v_{j,i}^2, \tag{5}$$

where $v_{j,i}$ is the ith entry of $\mathbf{v}_j$ and r is the rank of the right singular matrix (i.e., the number of principal components retained). This definition yields a nonnegative measure of a feature's total projection energy across leading directions. Features with large α_i thus either capture the core structural patterns of normal nodes or reflect the distributed deviations exploited by anomalies.

4.1.2. *Structure-aware feature selection*

Since real-world scenarios often deviate from the idealized SCM assumption, structural and edge anomalies may also play a crucial role in anomaly detection. Therefore, we fuse original features with aggregated neighborhood features to ensure that nodes with irregular neighborhood patterns can be captured in the spectral decomposition, thereby enhancing the practical applicability of our method:

$$\mathbf{X}' = (1-\beta)\mathbf{X} + \beta\mathbf{D}^{-1}\mathbf{A}\mathbf{X}, \tag{6}$$

where $\beta \in [0, 1]$ is a tunable hyperparameter and $\mathbf{D}$ is the degree diagonal matrix. We then perform SVD on $\mathbf{X}' \in R^{n\times f}$, obtaining $\mathbf{X}' = \mathbf{U}'\mathbf{\Sigma}'\mathbf{V}'^{\mathrm{T}}$. For each feature, we compute its spectral contribution $\boldsymbol{\alpha}_i$ using $\mathbf{V}'$, and construct the contribution vector $\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_f]$, we select $top_d = \underset{j\in 1,2,\ldots,f}{\operatorname{argmax}} \ \boldsymbol{\alpha}_j$ features with the highest contributions. The final selected feature matrix is constructed as:

$$\hat{\mathbf{X}} = \mathbf{X}[:, top_d], \tag{7}$$

This unsupervised strategy filters features based solely on spectral properties, without requiring label supervision, thereby avoiding the risk of label leakage.

4.2. Structure optimization based on neighborhood disparity

Recent studies such as GHRN (Gao et al., 2023a), which primarily focuses on edge deletion, and SparseGAD (Gong et al., 2023), which performs both edge addition and deletion, have demonstrated that structural optimization can substantially improve graph anomaly detection performance. These works collectively indicate that adapting the graph structure is not only feasible but also empirically effective. However, they generally treat all nodes in a uniform manner without explicitly considering the heterogeneity between head and tail nodes–a ubiquitous characteristic of real-world graphs that typically exhibit heavy-tailed degree distributions.

In this subsection, we propose a structure optimization strategy that explicitly differentiates between head and tail nodes. Specifically, we design a bi-directional edge-pruning mechanism for head nodes and a similarity-based neighborhood enhancement strategy for tail nodes. These two components jointly enable fine-grained structural adjustment. Moreover, our approach can be interpreted from the perspective of bias-variance trade-off in message passing: pruning noisy edges connected to head nodes reduces variance induced by redundant or inconsistent messages, while enhancing the neighborhoods of tail nodes alleviates high bias caused by insufficient information aggregation.

4.2.1. Bi-directional edge pruning for head nodes

Head nodes typically have many neighbors, increasing the risk of noisy or redundant connections that can degrade model's ability to aggregate informative signals. To address this, we propose a bi-directional pruning strategy based on average neighborhood similarity. This method removes low-quality edges while preserving important structural information. From a representation learning perspective, head nodes are particularly vulnerable to over-smoothing and over-squashing because aggregated messages come from a large number of heterogeneous neighbors. Low-similarity neighbors tend to inject conflicting signals that distort the embedding of head nodes. Therefore, selectively pruning such edges not only denoises local neighborhoods but also improves the signal-to-noise ratio during aggregation.

In this strategy, whether the edge $e_{i,j}$ should be removed depends not only on the pairwise similarity $sim_{i,j}$ between node v_i and v_j, but also on their respective average similarities with their neighbors. Specifically, if $sim_{i,j}$ is lower than both the average similarity of v_i to its neighbors (sim_{avg_i}) and that of v_j to its neighbors (sim_{avg_j}), the edge is considered to have low information value. Such edges are included in the candidate edge removal set $\mathcal{E}_c$, formally defined as:

$$\mathcal{E}_c = \left\{ (v_i, v_j) \in \mathcal{E} \;\middle|\; sim_{i,j} < sim_{avg_i},\ sim_{i,j} < sim_{avg_j},\ v_i \in \mathcal{V}_{head} \text{ or } v_j \in \mathcal{V}_{head} \right\}, \tag{8}$$

where cosine similarity is defined as $sim_{i,j} = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|}$, $\mathbf{x}_i = \hat{\mathbf{X}}[i, :]$ and the average neighborhood similarity is $sim_{avg_i} = \frac{1}{|\mathcal{N}(v_i)|} \sum_{v_j \in \mathcal{N}(v_i)} sim_{i,j}$.

To avoid indiscriminately removing all candidate edges, some of which may hold structural value, we adopt a probabilistic pruning strategy. Edges with lower similarity scores are assigned higher probabilities of removal. The probability $p_{i,j}$ for edge $(v_i, v_j) \in \mathcal{E}_c$ is defined as:

$$p_{i,j} = \frac{w_{i,j}}{\sum_{(v_k, v_l) \in \mathcal{E}_c} w_{k,l}}, \tag{9}$$

where $w_{i,j} = 1 - \sigma\left(\frac{sim'_{i,j} - 0.5}{\tau}\right)$ with $sim'_{i,j} = \frac{sim_{i,j} - \min\limits_{(v_k, v_l) \in \mathcal{E}_c} sim_{k,l}}{\max\limits_{(v_k, v_l) \in \mathcal{E}_c} sim_{k,l} - \min\limits_{(v_k, v_l) \in \mathcal{E}_c} sim_{k,l}}$, $\sigma(\cdot)$ is sigmoid function and τ is temperature hyperparameter controlling sensitivity.

4.2.2. Similarity-based neighborhood enhancement for tail nodes

Tail nodes suffer from limited connectivity, restricting information propagation and leading to weaker feature learning. To address this, we introduce a neighborhood enhancement strategy that connects tail nodes to structurally similar but currently unlinked nodes. Different from head nodes, tail nodes are typically affected by under-reaching, where insufficient neighborhood size prevents effective message passing. This may cause their embeddings to remain under-informed. Our enhancement strategy explicitly compensates for this by creating semantically meaningful connections based on feature similarity.

For each node $v_i \in \mathcal{V}_{tail}$, to enhance its neighborhood information while avoiding additional load on high-degree nodes, we construct a candidate neighbor set $\mathcal{V}_c$ by excluding $\mathcal{V}_{head} \cup \{v_i\}$ from the full node set $\mathcal{V}$. We then compute cosine similarity $sim_{i,c}$ between v_i and each $v_c \in \mathcal{V}_c$, and select the top $m = k\mu$ nodes with the highest similarity, where μ denotes the average node degree

in the graph, as augmented neighbor set. New edges are then created between v_i and these selected nodes and added to the graph. Formally defined as:

$$e_{i,c} \longleftarrow 1 \quad if \quad v_i \in Top_m\left(\left\{sim_{i,c} \mid v_c \in \mathcal{V}_c\right\}\right), \forall e_{i,c} = 0. \tag{10}$$

To ensure scalability, we randomly sample $3m$ nodes from $\mathcal{V}_c$ to form a reduced candidate set $\mathcal{V}_c'$. Similarities are computed only within $\mathcal{V}_c'$, from which the top-m nodes are selected for connection. This reduces computation while maintaining enhancement quality, making the strategy practical for large-scale graphs.

4.2.3. Final prediction

The optimized feature matrix $\hat{\mathbf{X}}$ and adjacency matrix $\hat{\mathbf{A}}$ are fed into an anomaly detection model $f(\cdot)$ to predict labels for unlabeled nodes $\hat{\mathbf{y}}_U = f\left(\hat{\mathbf{X}}, \hat{\mathbf{A}}; \boldsymbol{\theta}\right)$, where $\boldsymbol{\theta}$ denotes the model parameters and $\hat{\mathbf{y}}_U$ is the predicted anomaly score for the unlabeled set $\mathcal{V}_U$. The model is trained using a cross-entropy loss to optimize $\boldsymbol{\theta}$.

5. Novelty analysis

FSCAN, designed as a plugin to enhance downstream graph anomaly detection tasks, is the first to jointly optimize both raw graph features and structures. It can be seamlessly integrated into both unsupervised and supervised settings. Unlike existing approaches such as McGAD (Huang et al., 2026) and Noise-Resistant Model (Liu et al., 2025), which primarily focus on directly optimizing encoders and decoders, FSCAN directly improves the input graph quality, thereby strengthening the expressiveness of embeddings learned by any downstream detector. Theoretically, FSCAN introduces a spectral contribution-based feature selection strategy that identifies label-relevant dimensions from raw features, in contrast to existing feature optimization methods such as GDN (Gao et al., 2023b) and SEC-GFD (Xu et al., 2024), which rely on model performance gradient saliency to highlight important latent embeddings. Practically, this design ensures that anomaly-related signals are preserved at the data level before being encoded, providing a more principled way to mitigate feature redundancy. Moreover, unlike structure optimization approaches such as GHRN (Gao et al., 2023a) and SparseGAD (Gong et al., 2023), which refine graph connectivity based on model-learned embeddings, FSCAN leverages the spectrally filtered raw features to adaptively prune or add edges in the original topology. This data-driven rewiring strategy avoids model bias, guarantees broader applicability, and enables consistent improvements across both supervised and unsupervised anomaly detection tasks. Overall, FSCAN establishes a new paradigm of graph optimization as a plug-in, delivering both theoretical guarantees and empirical evidence for enhanced anomaly separability.

6. Experiments

In this section, we have conducted extensive experiments to evaluate the effectiveness of the proposed FSCAN.

6.1. Experimental details

6.1.1. Datasets and evaluation metrics

We evaluate FSCAN on five real-world GAD datasets: Amazon (Qiao et al., 2025), BlogCatalog (Luo et al., 2022), Questions (Dong et al., 2025) ,Cora (Luo et al., 2022), and OGB_arxiv (Hu et al., 2020), with diverse feature dimensions and domains. All datasets are summarized in Table 1 and publicly available at https://drive.google.com/drive/folders/1b_oh3VyOX2mODCR4Eg8zhV2durVkjbKf. Detailed descriptions are as follows:

Amazon is a product review dataset in the Musical Instruments category. Nodes represent users, and three types of edges are constructed: U-P-U connects users who reviewed the same product; U-S-U connects users who gave the same star rating; and U-V-U connects users who share the top 5% most similar reviews.

BlogCatalog is derived from a blog-sharing platform. Nodes represent users, and edges capture follower-followee relationships. Node attributes are based on tags associated with users.

Questions is collected from the Yandex Q&A platform. Nodes denote users, and an edge is formed between two users if one answers the other's question. Anomalies are defined as users who became inactive after August 2022.

Cora is a citation network dataset consisting of scientific publications in machine learning. Nodes correspond to papers, and edges represent citation relationships between them. Each node is associated with a sparse bag-of-words feature vector extracted from the paper's content, and each paper is labeled according to its research topic category.

OGB-arxiv is a large-scale paper citation network derived from the ARXIV repository. Nodes represent scientific papers, and directed edges denote citation relationships between papers. Each node is associated with a feature vector constructed from the paper's title and abstract using a bag-of-words representation, capturing its semantic content. Dataset statistics are presented in Table 1.

Given the inherent class imbalance in GAD, we adopt three widely used metrics for fair evaluation: PR-AUC, GMean, and F1. These metrics assess performance from different perspectives, especially in detecting minority classes. Higher scores indicate better anomaly detection performance.

Table 1
Dataset statistics.

Datasets	Type	#Nodes	#Edges	#Attributes	Anomalies (Rate)
Amazon	Co-Review	11,944	4,398,392	25	821 (6.87%)
Questions	Question Answering	48,921	153,540	301	1460 (2.98%)
BlogCatalog	Social Network	5196	171,743	8189	300 (5.77%)
Cora	Citation Networks	2,708	5,429	1,433	149 (5.5%)
OGB_arxiv	Citation Networks	169,343	1,166,243	128	16087 (9.5%)

6.1.2. Baselines

FSCAN is developed as an unsupervised plug-in that can be seamlessly incorporated into existing models. We first integrate it into representative GAD methods (GCN Kipf & Welling, 2016, BWGNN Tang et al., 2022) to demonstrate its compatibility and effectiveness. Beyond this, FSCAN is compared with the latest supervised feature optimization methods (GDN Gao et al., 2023b, SEC-GFD Xu et al., 2024) and supervised structure optimization methods (SparseGAD Gong et al., 2023, GHRN Gao et al., 2023a). To further validate its generality in the unsupervised setting, FSCAN is evaluated against method CoLA (Liu et al., 2022b) frameworks as well as the recent unsupervised structure optimization method AD-GCL (Xu et al., 2025b). We do not integrate FSCAN into other optimization baselines so as not to override their core designs.

6.1.3. Implementation details

We split each dataset into 40% training, 20% validation, and 40% test sets for all semi-supervised methods. We report mean and standard deviation over 10 runs. To ensure a fair comparison across all baselines, we adopt the following implementation strategies. For GCN, we re-implement the model using PyTorch. For all other baselines, we use their official open-source implementations, and set hyperparameters to the best configurations reported in the original papers. All baselines are evaluated under the same data splits across the three datasets.

6.2. Main results

To validate the effectiveness of FSCAN, we evaluate its performance against multiple baseline methods and conduct comprehensive ablation studies. Specifically, we investigate variants by removing the edge deletion module (FSCAN w/o del), the edge augmentation module (FSCAN w/o aug), the entire structural optimization module (FSCAN w/o str), and the feature selection module (FSCAN w/o feat). Results are summarized in Tables 2–4.

6.2.1. Supervised methods results

GNN models integrated with FSCAN consistently outperform their original versions across all three datasets. Among them, FSCAN(BWGNN) or FSCAN(GCN) achieves the best performance overall, showing that the joint optimization of feature selection and structure-aware refinement effectively improves anomaly detection. GDN separates node features into normal and abnormal modes, reducing redundancy and performing well on high-dimensional data. However, its extended version SEC-GFD performs worse on BlogCatalog, possibly due to increased redundancy from multi-frequency neighborhood fusion. Structure optimization methods like SparseGAD and GHRN outperform GDN on some datasets, confirming the importance of removing structural noise. Additionally, GHRN, which performs only edge deletion, achieves better results than SparseGAD, which applies both deletion and addition, suggesting that global edge modification may introduce new noise. Nevertheless, FSCAN consistently achieves the best performance across all datasets, further demonstrating its superior effectiveness. On the large-scale dataset OGB_arxiv, the performance of GCN ranks only second to FSCAN(GCN) and is clearly better than existing structure- and feature-optimization approaches. This indicates that, on large and complex graph datasets, a vanilla GCN already provides a strong representation ability, but it is still insufficient to fully mitigate the adverse effects of noisy structures and redundant features. In contrast, FSCAN, by jointly performing structure optimization and feature selection, substantially enhances representation quality, further unlocking the potential of GCN and yielding more pronounced performance gains in complex real-world scenarios.

6.2.2. Unsupervised methods results

On Amazon and BlogCatalog, FSCAN (CoLA) significantly outperforms AD-GCL, demonstrating its effectiveness and stability in enhancing anomaly detection. On BlogCatalog and Cora datasets, AD-GCL yields a GMean of 0 due to completely misclassifying all anomalies. On the Questions, AD-GCL failed to run due to memory limitations. Meanwhile, FSCAN (CoLA) performs slightly worse than CoLA in terms of PR-AUC and F1. This may stem from the extremely sparse structure of the Questions graph, where even slight perturbations can disrupt local neighborhoods. As CoLA heavily relies on subgraph sampling to capture contextual information, such structural fragility undermines CoLA's detection performance. Nevertheless, FSCAN still improves GMean, validating the core mechanism. Moreover, on the OGB-arixv dataset, the large number of nodes and edges leads to an excessive number of subgraph constructions, causing all unsupervised methods to encounter out-of-memory (OOM) issues.

6.2.3. Component effectiveness analysis.

Based on Tables 1 and 2, removing either structural optimization component results in varying degrees of performance drop, indicating that both edge deletion and edge enhancement are essential for improving anomaly detection performance.

Table 2
Performance comparison of different methods on Amazon and Questions datasets.

Methods	Amazon			Questions		
	F1	PR-AUC	GMean	F1	PR-AUC	GMean
GCN	72.33 ± 0.40	42.55 ± 6.23	72.87 ± 1.45	57.96 ± 0.46	13.67 ± 0.50	41.60 ± 1.98
BWGNN	91.92 ± 0.59	88.53 ± 2.42	90.65 ± 1.09	60.03 ± 1.31	15.47 ± 1.58	40.81 ± 2.58
GDN	90.28 ± 0.13	86.89 ± 0.22	88.46 ± 0.07	37.72 ± 0.40	5.11 ± 0.23	48.26 ± 0.57
SEC-GFD	90.85 ± 0.55	87.45 ± 0.65	87.71 ± 0.71	58.14 ± 2.26	11.21 ± 0.94	43.22 ± 4.64
SparseGAD	90.64 ± 0.52	86.96 ± 1.57	85.10 ± 0.93	54.83 ± 0.68	15.51 ± 1.40	24.46 ± 1.59
GHRN	91.61 ± 0.78	89.20 ± 1.09	90.05 ± 1.10	59.31 ± 0.52	12.86 ± 0.56	46.60 ± 1.69
FSCAN(GCN)	74.68 ± 1.92	52.13 ± 5.18	78.20 ± 0.85	57.84 ± 0.51	13.18 ± 0.30	**58.51 ± 0.57**
FSCAN(BWGNN)	**92.61 ± 0.35**	**90.37 ± 0.33**	**91.56 ± 0.40**	**60.12 ± 0.44**	**16.22 ± 0.97**	45.41 ± 1.62
CoLA	51.52 ± 0.13	6.99 ± 0.03	21.39 ± 6.19	**59.79 ± 0.19**	**3.86 ± 0.03**	20.85 ± 2.11
AD-GCL	53.24 ± 0.14	7.06 ± 0.03	21.02 ± 6.75	OOM	OOM	OOM
CoLA + FSCAN	**58.04 ± 0.13**	**7.78 ± 0.04**	**31.03 ± 0.55**	54.69 ± 0.14	3.74 ± 0.01	**24.86 ± 0.25**
FSCAN-w/o del	92.48 ± 0.60	90.30 ± 0.44	91.23 ± 0.42	58.51 ± 0.57	13.87 ± 1.25	47.04 ± 1.16
FSCAN-w/o aug	92.60 ± 0.36	90.36 ± 0.52	91.44 ± 0.50	58.70 ± 0.39	13.17 ± 0.55	46.78 ± 1.06
FSCAN-w/o str	91.98 ± 0.54	90.38 ± 0.38	91.87 ± 0.63	59.55 ± 0.57	15.60 ± 0.86	45.21 ± 1.16
FSCAN-w/o feat	91.46 ± 0.41	84.95 ± 1.48	90.79 ± 0.53	58.99 ± 0.54	14.46 ± 0.78	46.01 ± 1.77

Table 3
Performance comparison of different methods on BlogCatalog and OGB_arxiv datasets.

Methods	BlogCatalog			OGB_arxiv		
	F1	GMean	PR-AUC	F1	PR-AUC	GMean
GCN	64.09 ± 0.80	28.26 ± 0.3	48.32 ± 1.94	74.01 ± 0.30	46.93 ± 1.17	69.97 ± 0.75
BWGNN	69.62 ± 2.30	31.28 ± 1.94	58.14 ± 3.00	59.95 ± 0.35	16.60 ± 1.03	50.36 ± 1.15
GDN	47.24 ± 4.62	**38.06 ± 1.66**	60.72 ± 0.84	47.44 ± 0.04	41.76 ± 0.67	66 ± 0.19
SEC-GFD	47.85 ± 1.91	11.49 ± 1.32	42.44 ± 1.64	65.99 ± 0.22	28.02 ± 0.74	56.87 ± 2.03
SparseGAD	61.85 ± 2.76	21.78 ± 4.73	50.90 ± 3.19	62.66 ± 3.45	22.33 ± 6.36	51.95 ± 4.91
GHRN	69.30 ± 1.08	31.40 ± 0.59	59.25 ± 3.22	57.44 ± 0.55	11.83 ± 0.58	52.77 ± 2.74
FSCAN(GCN)	65.38 ± 0.39	30.93 ± 0.44	50.13 ± 0.56	**81.04 ± 0.39**	**68.10 ± 0.48**	77.49 ± 0.85
FSCAN(BWGNN)	**72.37 ± 0.67**	37.96 ± 1.31	**67.35 ± 1.07**	72.18 ± 1.76	40.35 ± 3.34	**78.10 ± 2.09**
CoLA	77.79 ± 0.32	33.68 ± 0.52	11.51 ± 0.92	OOM	OOM	OOM
AD-GCL	71.60 ± 0.16	23.25 ± 0.18	0.00	OOM	OOM	OOM
CoLA + FSCAN	**78.32 ± 0.29**	**36.05 ± 0.29**	**22.49 ± 1.03**	OOM	OOM	OOM
FSCAN-w/o del	70.87 ± 0.92	37.99 ± 2.33	67.89 ± 1.84	64.94 ± 0.35	25.04 ± 0.92	60.80 ± 1.02
FSCAN-w/o aug	71.62 ± 0.98	36.95 ± 1.65	66.60 ± 1.91	71.90 ± 1.92	40.00 ± 4.08	77.89 ± 1.99
FSCAN-w/o str	72.19 ± 0.59	37.63 ± 1.27	66.67 ± 0.91	72.10 ± 1.58	40.27 ± 2.77	77.85 ± 2.59
FSCAN-w/o feat	71.73 ± 0.70	35.55 ± 1.12	66.70 ± 1.43	65.41 ± 0.31	23.51 ± 0.41	59.80 ± 0.69

6.3. Effectiveness analysis of feature selection

To assess the effectiveness of SC, we compare it with two supervised baselines: RF (Buchaiah & Shakya, 2022b) and TFI. Experiments are conducted on three real-world datasets with varying feature dimensions. We rank feature importance for each method and select the top 10% to 100% of features. These subsets are input to BWGNN and evaluated on the test set using F1 and PR-AUC scores (Fig. 5).

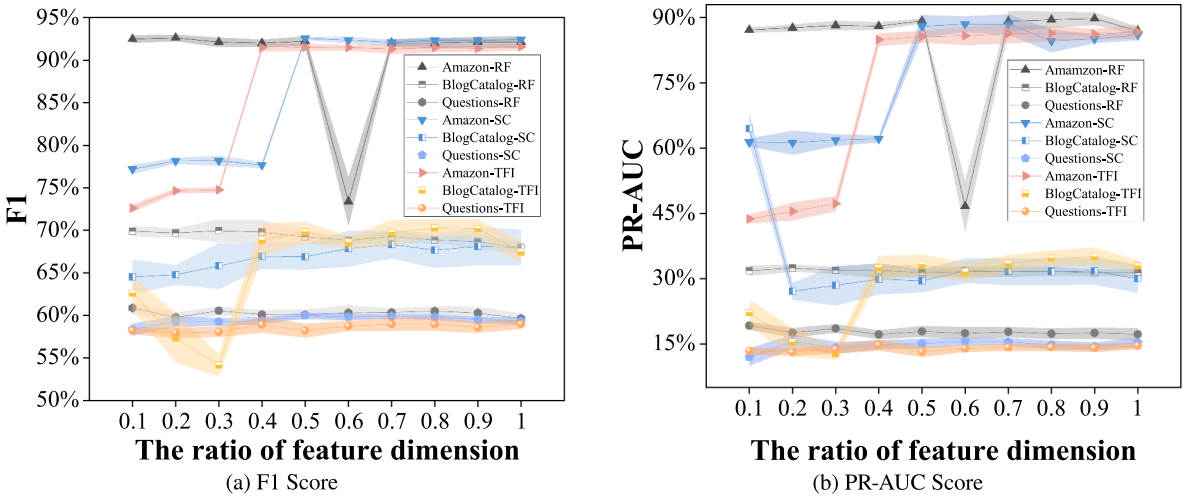
Fig. 5 reveals several key insights. First, using all features does not yield optimal performance–F1 and PR-AUC scores exhibit a clear non-monotonic trend as the number of selected features increases. This indicates that the original feature space contains redundant and noisy information. As feature dimensionality grows, the accumulation of noise from redundant features becomes more pronounced. This is especially evident on the high-dimensional BlogCatalog dataset, where feature selection leads to the most significant performance gains. Even on the mid-dimensional Questions dataset, where model performance remains relatively stable, feature selection still provides noticeable improvements, highlighting the ubiquity of feature redundancy.

Second, when comparing the two selection strategies, SC consistently demonstrates greater stability across all datasets. In high-dimensional settings, the feature subspace constructed by SC offers better separability for anomalous patterns. In medium and low-dimensional cases, while performance differences between methods are smaller, SC exhibits smoother trends across varying feature ratios, indicating more robust behavior.

Lastly, from a methodological perspective, both RF and TFI rely on supervised signals to estimate feature importance, making them susceptible to biases introduced by label quality or training/test splits. In contrast, SC is an unsupervised approach that eval-

Table 4
Performance comparison of different methods on Cora dataset.

Methods	Cora		
	F1	PR-AUC	GMean
GCN	80.08 ± 1.76	60.74 ± 1.07	75.10 ± 0.70
BWGNN	65.72 ± 4.37	30.34 ± 8.23	53.34 ± 7.43
GDN	51.75 ± 4.85	37.06 ± 1.56	61.76 ± 1.88
SEC-GFD	50.80 ± 2.28	9.16 ± 2.21	31.30 ± 5.65
SparseGAD	54.10 ± 1.86	10.22 ± 2.54	28.82 ± 5.65
GHRN	59.43 ± 2.30	15.09 ± 2.82	50.64 ± 6.33
FSCAN(GCN)	**84.00 ± 0.86**	**63.62 ± 1.18**	**76.16 ± 0.09**
FSCAN(BWGNN)	68.19 ± 2.56	33.52 ± 2.81	75.03 ± 1.42
CoLA	90.23 ± 0.27	**48.26 ± 1.34**	28.30 ± 2.14
AD-GCL	**90.37 ± 0.31**	41.94 ± 1.00	0.00 ± 0.00
CoLA + FSCAN	88.67 ± 0.38	45.19 ± 1.14	**33.89 ± 1.34**
FSCAN-w/o del	69.13 ± 1.55	34.31 ± 1.67	75.06 ± 2.62
FSCAN-w/o aug	67.01 ± 1.97	32.02 ± 5.96	74.78 ± 2.07
FSCAN-w/o str	66.99 ± 1.87	31.64 ± 3.75	72.64 ± 3.59
FSCAN-w/o feat	68.19 ± 2.56	33.52 ± 2.81	75.03 ± 1.42



(a) F1 Score (b) PR-AUC Score

Fig. 5. Variation of F1 and PR-AUC scores on the test sets of different datasets using different feature selection strategies.

uates features based on their representation capacity in the spectral space. As it does not require label information, SC offers better generalizability and practical applicability.

6.4. Effectiveness of structural optimization strategy

Beyond feature selection, the structure optimization strategy based on neighborhood differences is a core component of FSCAN. To assess its role in mitigating performance disparities between head and tail nodes, we visualize the mean and variance of F1 using bar charts with error bars for BWGNN and BWGNN + FSCAN across three benchmark datasets, as shown in Fig. 6.

In tail nodes analysis (Fig. 6(a)), FSCAN shows a slight performance drop compared to the original BWGNN on the BlogCatalog. This can be attributed to the high-dimensional feature space of BlogCatalog. Even after feature selection, a large number of redundant dimensions remain, reducing the accuracy of similarity calculations. As a result, the similarity-based edge augmentation may introduce semantically irrelevant, disrupting local structural consistency and weakening the model's ability to represent tail nodes. Despite this decline on BlogCatalog, FSCAN demonstrates stable or slightly improved tail node performance on the Questions and Amazon, indicating the adaptability of the proposed similarity-based edge addition strategy. This strategy enhances the structural awareness of low-degree nodes while avoiding significant disruption to original graph topology.

For head nodes (Fig. 6(b)), FSCAN yields notable performance improvements over original BWGNN. It either stabilizes performance by reducing variance or boosts it by improving F1. This suggests that bi-directional edge deletion strategy effectively reduces structural noise caused by redundant or overly dense neighborhoods, thereby enhancing the discriminative power of head node representations.

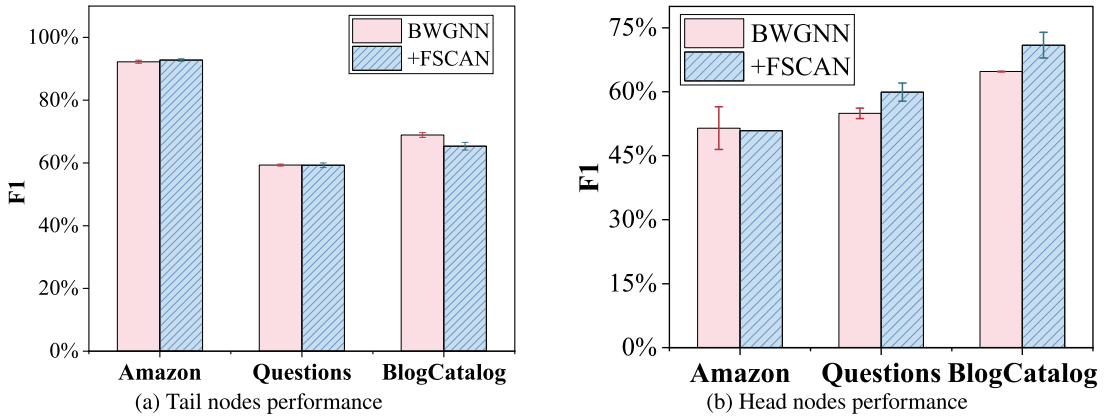


Fig. 6. Detection performance on head and tail nodes using BWGNN and BWGNN with FSCAN.

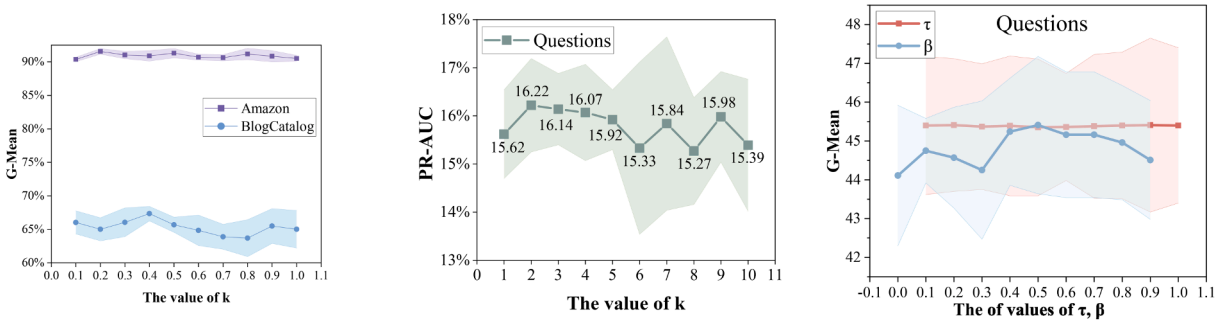


Fig. 7. Impact of hyperparameters k, τ, and β on performance.

6.5. Hyperparameter analysis

To evaluate the impact of key hyperparameters k, β, and τ on model performance, we conduct a comprehensive sensitivity analysis of these parameters.

The hyperparameter k controls the number of top m neighboring nodes selected during the tail-node neighborhood enhancement stage.We analyze its effect on the Amazon, Questions, and BlogCatalog datasets, and the corresponding results are reported in the first two subfigures of Fig. 7. As shown, the Amazon dataset exhibits low sensitivity to the hyperparameter k, with only marginal performance variations. In contrast, the Questions and BlogCatalog datasets are more sensitive to this parameter, showing noticeable performance changes as k varies. This effect is particularly pronounced on the Questions dataset, where the average node degree is low and the graph structure is relatively sparse. In such cases, the number of newly added neighbors plays a critical role in representation learning, thereby amplifying the model's sensitivity to k.

Furthermore, we analyze the influence of the feature–structure awareness weight β and the temperature parameter τ used in the head-node edge removal process on the Questions dataset. As illustrated in the last subfigure of Fig. 7, the model performance is more sensitive to β than to τ. When β is too small, structural information cannot be effectively incorporated into feature representations, whereas excessively large β may introduce noisy structural signals and degrade performance. In comparison, the model remains relatively robust to variations in τ, maintaining stable performance across a wide range of values.

6.6. Complexity analysis

We further analyze the time and space complexity of each module in FSCAN. The overall pipeline of FSCAN consists of three stages: feature selection, structure refinement (edge removal for head nodes and edge addition for tail nodes), and graph representation learning.

In the feature selection stage, FSCAN applies a randomized truncated SVD to the original feature matrix $\mathbf{X} \in R^{n \times f}$ to extract its dominant feature directions. This preprocessing step runs only once before training, with a reduced time complexity of $O(nfk \log k)$, where $k \ll \min(n, f)$. The moderate cost is acceptable as it does not affect subsequent iterative training. In the structure refinement stage, the mean similarity over the entire graph is first computed with a time complexity of $O(|\mathcal{E}|f)$. For the head node set $\mathcal{V}_{\text{head}}$, edge removal relies on similarity computations between nodes and their neighbors, resulting in a complexity of $O(|\mathcal{E}_{\mathcal{V}_{\text{head}}}|f)$. For tail nodes, similarity matching within a candidate set followed by edge addition incurs a complexity of $O(|\mathcal{E}_{3m}|f)$. Since $|\mathcal{E}|f$, $|\mathcal{E}_{\mathcal{V}_{\text{head}}}|f$,

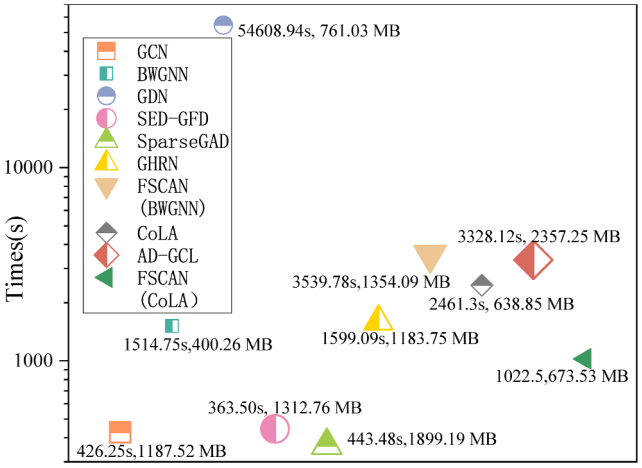


Fig. 8. Time and space complexity analysis on Amazon dataset.

Table 5
Runtime comparison of feature selection methods (in seconds).

Methods	SC	TFI	RF
Amazon	0.12	0.58	0.32
Questions	0.25	24.34	47.24
BlogCatalog	30.17	62.71	6.27

and $|\mathcal{E}_{3m}|f$ are all significantly smaller than n^2, the cost of structure refinement is substantially lower than that of structure learning methods requiring reconstruction over all node pairs, leading to better scalability. For graph representation learning on the refined graph, the time complexity is $O(af + alh)$, where a denotes the number of non-zero elements in the adjacency matrix, h is the hidden dimension, and l is the number of convolution layers. Overall, the total time complexity of FSCAN is $O(nfk\log k + |\mathcal{E}|f + |\mathcal{E}_{\mathcal{V}_{\text{head}}}|f + |\mathcal{E}_{3m}|f + af + alh)$.

To provide an empirical comparison of computational efficiency, Fig. 8 reports the runtime and memory consumption of FSCAN and baseline methods over 10 runs on the Amazon dataset. In Fig. 8, the size of each marker reflects the relative memory usage of the corresponding method. As shown in Fig. 8, under the semi-supervised setting, FSCAN (BWGNN) exhibits relatively higher time and memory costs. This is mainly due to the high space complexity of the spectral-based feature selection method SC and the additional similarity computations introduced by the structure refinement process. Nevertheless, FSCAN remains more efficient than GDN, which relies on model gradients for feature regularization and incurs higher computational overhead. Under the unsupervised setting, since edge removal dominates edge addition during structure refinement, the resulting graph becomes sparser. This improves the efficiency of subgraph computation and message passing, leading to reduced runtime. These results indicate that FSCAN maintains strong performance while achieving improved computational efficiency in unsupervised scenarios.

To further evaluate the efficiency of the SC feature selection method, we compare the runtime of SC, TFI, and RF across three datasets with substantially different feature dimensionalities, as reported in Table 5. As shown in Table 5, when the feature dimensionality is low, the runtime differences among the three methods are marginal. As the feature dimensionality increases, the computational cost of all methods grows accordingly. Since RF is a model-based feature selection method, its runtime depends not only on the number of features but also on the scale of the input graph, leading to significantly higher computational overhead on large graphs. In contrast, both TFI and SC rely on feature-level statistics, and their computational complexity mainly depends on feature dimensionality. As the number of features increases from Amazon to Questions and further to BlogCatalog, ranging from tens to hundreds and then to thousands, TFI exhibits a larger runtime increase, indicating that SC achieves better scalability in high-dimensional feature settings.

6.7. Visualization

To intuitively analyze the impact of different methods on node representations, we present t-SNE visualizations of the original features, FSCAN (BWGNN), the structure optimization method GHRN, and the feature optimization method SEC-GFD on the Questions dataset, as shown in Fig. 9.

As shown in Fig. 9, compared with the original features, all optimization-based methods produce embeddings with stronger discriminative ability, indicating that both feature-level and structure-level optimization contribute to improving the quality of node representations. In addition, the embeddings generated by FSCAN (BWGNN) form more compact clusters in the embedding space,

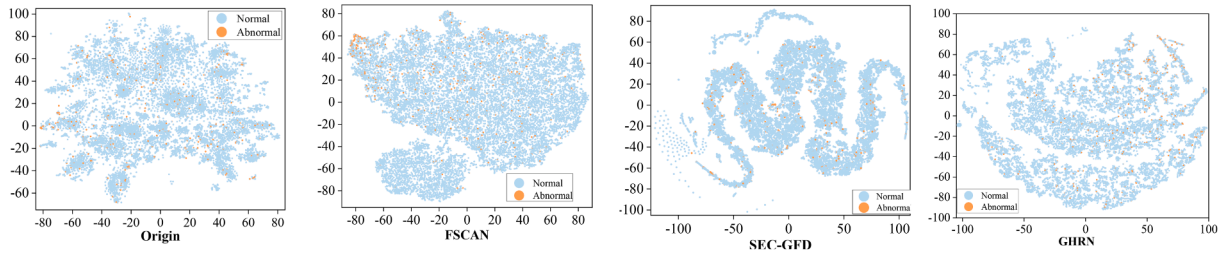


Fig. 9. t-SNE visualization of different methods on Questions dataset.

suggesting that the joint strategy of eliminating redundant features and optimizing graph structure is more conducive to the effective detection of anomalous nodes.

7. Conclusion

We propose FSCAN, a plug-in framework for improving graph anomaly detection. To tackle feature redundancy and structural noise in real-world graphs, FSCAN integrates spectral contribution-based feature selection with differentiated structure refinement to enhance representation quality. Experiments on three real-world datasets with diverse feature distributions show consistent gains over state-of-the-art baselines, validating its effectiveness. However, due to the high space complexity of feature selection and the substantial time complexity of structure refinement, FSCAN exhibits relatively high computational and memory overhead on large-scale graphs. In the future, efficiency could be further improved through approximate computations or incremental update strategies, enhancing the scalability of the framework in large-scale scenarios.

CRediT authorship contribution statement

Yingyue Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Huifang Ma:** Writing – review & editing; **Rui Bing:** Writing – review & editing, Supervision; **Meihuizi Jia:** Writing – review & editing, Supervision; **Xiaofei Wang:** Writing – review & editing.

Data availability

I have shared the link to my data/code at the Attach File step.

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References

Benaych-Georges, F., & Nadakuditi, R. R. (2012). The singular values and vectors of low rank perturbations of large rectangular random matrices. *Journal of Multivariate Analysis*, *111*, 120–135. https://doi.org/10.1016/j.jmva.2012.04.019

Bi, W., Du, L., Fu, Q., Wang, Y., Han, S., & Zhang, D. (2024). Make heterophilic graphs better fit GNN: A graph rewiring approach. *IEEE Transactions on Knowledge and Data Engineering*, *36* (12), 8744–8757. https://doi.org/10.1109/TKDE.2024.3441766

Buchaiah, S., & Shakya, P. (2022a). Bearing fault diagnosis and prognosis using data fusion based feature extraction and feature selection. *Measurement*, *188*, 110506. https://doi.org/10.1016/j.measurement.2021.110506

Buchaiah, S., & Shakya, P. (2022b). Bearing fault diagnosis and prognosis using data fusion based feature extraction and feature selection. *Measurement*, *188*, 110506. https://doi.org/10.1016/j.measurement.2021.110506

Chaudhuri, A. (2024). Search space division method for wrapper feature selection on high-dimensional data classification. *Knowledge-Based Systems*, *291*, 111578. https://doi.org/10.1016/j.knosys.2024.111578

Ding, X., & Yang, F. (2021). Spiked separable covariance matrices and principal components. *The Annals of Statistics*, *49* (2), 1113 – 1138. https://doi.org/10.1214/20-AOS1995

Gao, Y., Fang, J., Sui, Y., Li, Y., Wang, X., Feng, H., & Zhang, Y. (2024). Graph anomaly detection with bi-level optimization. In *Proceedings of the ACM web conference 2024* WWW '24 (p. 4383-4394). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3589334.3645673

Gao, Y., Wang, X., He, X., Liu, Z., Feng, H., & Zhang, Y. (2023a). Addressing heterophily in graph anomaly detection: A perspective of graph spectrum. In *Proceedings of the ACM web conference 2023* WWW '23 (p. 1528-1538). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3543507.3583268

Gao, Y., Wang, X., He, X., Liu, Z., Feng, H., & Zhang, Y. (2023b). Alleviating structural distribution shift in graph anomaly detection. In *Proceedings of the ACM international conference on web search and data mining* WSDM '23 (p. 357-365). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3539597.3570377

Gong, H., Li, Y., Zhang, J., Zhang, B., & Wang, X. (2024). A new filter feature selection algorithm for classification task by ensembling pearson correlation coefficient and mutual information. *Engineering Applications of Artificial Intelligence 131*, 107865. https://doi.org/10.1016/j.engappai.2024.107865

Gong, Z., Wang, G., Sun, Y., Liu, Q., Ning, Y., Xiong, H., & Peng, J. (2023). Beyond homophily: Robust graph anomaly detection via neural sparsification. In E. Elkind (Ed.), *Proceedings of the international joint conference on artificial intelligence, IJCAI'23* (pp. 2104–2113). International Joint Conferences on Artificial Intelligence. https://doi.org/10.24963/ijcai.2023/234

Hassan, A., Paik, J. H., Khare, S. R., & Hassan, S. A. (2025). A wrapper feature selection approach using markov blankets. *Pattern Recognition*, *158*, 111069. https://doi.org/10.1016/j.patcog.2024.111069

Hu, W., Fey, M., Zitnik, M., Dong, Y., Ren, H., Liu, B., Catasta, M., & Leskovec, J. (2020). Open graph benchmark: Datasets for machine learning on graphs. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems* (pp. 22118–22133). Curran Associates, Inc. (*vol. 33*). https://proceedings.neurips.cc/paper_files/paper/2020/file/fb60d411a5c5b72b2e7d3527cfc84fd0-Paper.pdf.

Huang, T., Wang, Y., Li, Q., & Gao, J. (2026). Multi-faceted consistency data augmentation for graph anomaly detection. *Information Processing & Management*, *63* (1), 104338. https://doi.org/10.1016/j.ipm.2025.104338

Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. arXiv:1609.02907

Li, K., Liu, Y., Ao, X., Chi, J., Feng, J., Yang, H., & He, Q. (2022). Reliable representations make a stronger defender: Unsupervised structure refinement for robust GNN. In *Proceedings of the ACM SIGKDD conference on knowledge discovery and data mining* KDD '22 (p. 925-935). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3534678.3539484

Li, Z., Wang, L., Sun, X., Luo, Y., Zhu, Y., Chen, D., Luo, Y., Zhou, X., Liu, Q., Wu, S., Wang, L., & Yu, J. (2023). GSLB: The graph structure learning benchmark: The graph structure learning benchmark. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in neural information processing systems* (pp. 30306–30318). Curran Associates, Inc. (*vol. 36*). https://proceedings.neurips.cc/paper_files/paper/2023/file/60bc87f3cf5257579435d92ec12c761b-Paper-Datasets_and_Benchmarks.pdf.

Liu, N., Wang, X., Wu, L., Chen, Y., Guo, X., & Shi, C. (2022a). Compact graph structure learning via mutual information compression. In *In proceedings of the ACM web conference 2022* WWW '22 (p. 1601-1610). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3485447.3512206

Liu, Y., Li, Z., Pan, S., Gong, C., Zhou, C., & Karypis, G. (2022b). Anomaly detection on attributed networks via contrastive self-supervised learning. *IEEE Transactions on Neural Networks and Learning Systems*, *33* (6), 2378–2392. https://doi.org/10.1109/TNNLS.2021.3068344

Liu, Y., Zheng, Y., Zhang, D., Chen, H., Peng, H., & Pan, S. (2022c). Towards unsupervised deep graph structure learning. In *In proceedings of the ACM web conference 2022* WWW '22 (p. 1392-1403). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3485447.3512186

Liu, Z., Nguyen, T.-K., & Fang, Y. (2021). Tail-GNN: Tail-node graph neural networks. In *In proceedings of the ACM SIGKDD conference on knowledge discovery and data mining* KDD '21 (p. 1109-1119). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3447548.3467276

Liu, Z., Yu, H., & Luo, X. (2025). A noise-resistant model for graph-based fraud detection. *Information Processing & Management*, *62* (5), 104198. https://doi.org/10.1016/j.ipm.2025.104198

Luo, X., Wu, J., Beheshti, A., Yang, J., Zhang, X., Wang, Y., & Xue, S. (2022). ComGA: Community-aware attributed graph anomaly detection. In *Proceedings of the ACM international conference on web search and data mining* WSDM '22 (p. 657-665). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3488560.3498389

Qiao, H., & Pang, G. (2023). Truncated affinity maximization: One-class homophily modeling for graph anomaly detection. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in neural information processing systems* (pp. 49490–49512). Curran Associates, Inc. (*vol. 36*). https://proceedings.neurips.cc/paper_files/paper/2023/file/9b905031125e56a557db38dff4fa8d21-Paper-Conference.pdf.

Tang, J., Li, J., Gao, Z., & Li, J. (2022). Rethinking graph neural networks for anomaly detection. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvari, G. Niu, & S. Sabato (Eds.), *Proceedings of the international conference on machine learning* (pp. 21076–21089). PMLR (*vol. 162*). Proceedings of Machine Learning Research. https://proceedings.mlr.press/v162/tang22b.html.

Wang, R., Mou, S., Wang, X., Xiao, W., Ju, Q., Shi, C., & Xie, X. (2021). Graph structure estimation neural networks. In *Proceedings of the web conference 2021* WWW '21 (p. 342-353). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3442381.3449952

Wu, L., Lin, H., Liu, Z., Liu, Z., Huang, Y., & Li, S. Z. (2024). Homophily-enhanced self-supervision for graph structure learning: Insights and directions. *IEEE Transactions on Neural Networks and Learning Systems* , *35* (9), 12358–12372. https://doi.org/10.1109/TNNLS.2023.3257325

Wu, Q., Zhao, W., Li, Z., Wipf, D. P., & Yan, J. (2022). NodeFormer: A scalable graph structure learning transformer for node classification. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, & A. Oh (Eds.), *Advances in neural information processing systems* (pp. 27387–27401). Curran Associates, Inc. (*vol. 35*). https://proceedings.neurips.cc/paper_files/paper/2022/file/af790b7ae573771689438bbcfc5933fe-Paper-Conference.pdf.

Xu, F., Wang, N., Wu, H., Wen, X., Zhao, X., & Wan, H. (2024). Revisiting graph-based fraud detection in sight of heterophily and spectrum. *In Proceedings of the AAAI Conference on Artificial Intelligence*, *38* (8), 9214–9222. https://doi.org/10.1609/aaai.v38i8.28773

Xu, X., Wei, F., Yu, T., Lu, J., Liu, A., Zhuo, L., Nie, F., & Wu, X. (2025a). Embedded multi-label feature selection via orthogonal regression. *Pattern Recognition*, *163*, 111477. https://doi.org/https://doi.org/10.1016/j.patcog.2025.111477

Xu, Y., Peng, Z., Shi, B., Hua, X., Dong, B., Wang, S., & Chen, C. (2025b). Revisiting graph contrastive learning on anomaly detection: A structural imbalance perspective. *In Proceedings of the AAAI Conference on Artificial Intelligence*, *39* (12), 12972–12980. https://doi.org/10.1609/aaai.v39i12.33415

Zhang, G., Yuan, G., Cheng, D., Liu, L., Li, J., & Zhang, S. (2025). Disentangled contrastive learning for fair graph representations. *Neural Networks*, *181*, 106781. https://doi.org/10.1016/j.neunet.2024.106781

Zhang, Y., Sharma, K., Du, L., & Liu, Y. (2024). Toward mitigating misinformation and social media manipulation in LLM era. In *Proceedings of the ACM web conference 2024* WWW '24 (p. 1302-1305). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3589335.3641256

Zhao, J., Wang, X., Shi, C., Hu, B., Song, G., & Ye, Y. (2021). Heterogeneous graph structure learning for graph neural networks. *In Proceedings of the AAAI Conference on Artificial Intelligence*, *35* (5), 4697–4705. https://doi.org/10.1609/aaai.v35i5.16600

Zhao, J., Wen, Q., Ju, M., Zhang, C., & Ye, Y. (2023). Self-supervised graph structure refinement for graph neural networks. In *Proceedings of the ACM international conference on web search and data mining* WSDM '23 (p. 159-167). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3539597.3570455

Zheng, Y., Li, X., Luan, S., Peng, X., & Chen, L. (2025). Let your features tell the differences: Understanding graph convolution by feature splitting. In *The international conference on learning representations* (pp. 1–15). https://openreview.net/forum?id=I9omfcWfMp.

Zhou, Z., Zhou, S., Mao, B., Zhou, X., Chen, J., Tan, Q., Zha, D., Feng, Y., Chen, C., & Wang, C. (2023). OpenGSL: A comprehensive benchmark for graph structure learning. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in neural information processing systems* (pp. 17904–17928). Curran Associates, Inc. (*vol. 36*). https://proceedings.neurips.cc/paper_files/paper/2023/file/39f8ef62e061042cca8c8f46d7e0e31b-Paper-Datasets_and_Benchmarks.pdf.